Predicting In-Hospital Mortality Using Machine Learning on the MIMIC-III Clinical Database

1. Introduction

In-hospital mortality is an important clinical outcome that can be used to assess patient outcomes and investigate factors associated with severe illness. Identifying patients who may be at an increased risk of death is a potential application of predictive analytics in healthcare. Machine learning techniques can analyse patterns across demographic characteristics, laboratory measurements and other clinical variables to estimate the likelihood of a particular outcome. However, the usefulness of such models depends on the quality and quantity of the available data, the selection of appropriate predictors and the reliability of model evaluation.

This project investigates the use of machine learning to predict in-hospital mortality using data from the Medical Information Mart for Intensive Care III (MIMIC-III) clinical database (Johnson et al., 2016). The analysis focuses on information associated with hospital admissions, including patient age, sex, laboratory measurements and diagnostic categories. The intended prediction task is to distinguish between admissions that resulted in in-hospital death and those that did not.

The central research question is: Can machine learning models predict in-hospital mortality using information available during the first 24 hours of a hospital admission? To investigate this question, the project develops and compares several classification models, including Logistic Regression, Random Forest and XGBoost. Their performance is assessed using cross-validation and a held-out test set, with particular attention to mortality recall and other classification metrics.

The project uses the MIMIC-III Clinical Database Demo, a limited demonstration dataset. Consequently, the analysis is intended as a proof of concept rather than the development of a clinically deployable prediction tool. The small sample size limits the reliability and generalisability of the results, and the findings must therefore be interpreted cautiously. The project aims to demonstrate a structured healthcare data science workflow, from data preparation and exploratory analysis to model development, evaluation and critical interpretation.

2. Methods

2.1 Dataset

This project uses the MIMIC-III Clinical Database Demo, a demonstration version of the Medical Information Mart for Intensive Care III database. The dataset contains de-identified clinical information relating to hospital admissions and patient characteristics. The demonstration dataset was selected to develop and evaluate a machine learning workflow for predicting in-hospital mortality (Johnson et al., 2016; Johnson, Pollard and Mark, 2019).

The analysis includes 129 hospital admissions from 100 unique patients. Each observation represents a hospital admission rather than a unique individual, allowing the outcome to be defined at the admission level. The target variable is **hospital_expire_flag**, a binary indicator recording whether the patient died during the hospital admission. A value of 1 indicates in-hospital death, while a value of 0 indicates that the patient survived to hospital discharge.

Of the 129 admissions, 40 (31.0%) resulted in in-hospital death and 89 (69.0%) did not. This distribution indicates a moderate class imbalance, with surviving admissions forming the majority class. Consequently, model performance was assessed using metrics beyond accuracy alone, including recall for the mortality class.

The selected predictors comprise **age at admission, gender, diagnostic category** and 13 laboratory measurements: **bicarbonate, calcium, creatinine, glucose, magnesium, potassium, sodium, urea nitrogen, haematocrit, haemoglobin, platelet count, red cell distribution width (RDW)** and **white blood cell count**. These variables were selected to explore whether demographic, diagnostic and laboratory information could support mortality prediction.

2.2 Data Cleaning and Preprocessing

The raw MIMIC-III tables were inspected and processed to construct an admission-level dataset for mortality prediction. Relevant information was extracted from the admissions, patients, laboratory events, laboratory item dictionary and diagnosis tables. Admission identifiers were used to combine information from these sources, with the aim of retaining one observation per hospital admission (Johnson et al., 2016).

Thirteen laboratory variables were selected for analysis: **bicarbonate, calcium, creatinine, glucose, magnesium, potassium, sodium, urea nitrogen, haematocrit, haemoglobin, platelet count, red cell distribution width** and **white blood cell count**. Laboratory event timestamps were compared with admission times to calculate the elapsed time between admission and each measurement. Only measurements recorded between admission and 24 hours after admission were retained for predictor construction.

For admissions with multiple measurements of the same laboratory test during this period, the most recent measurement was selected. This produced a single value for each available laboratory variable within each admission, avoiding the need to treat repeated measurements as separate observations. Measurements timestamped before admission were investigated separately and excluded from the final 24-hour laboratory selection.

Patient age at admission was calculated using the available date-of-birth and admission-date information. Gender and diagnostic information were also incorporated into the dataset. Diagnostic categories were derived from the diagnosis data, with categories occurring fewer than five times grouped into an "**Other**" category to reduce the number of infrequently represented groups. The binary outcome, **hospital_expire_flag**, was retained as the target variable, with 1 representing in-hospital death and 0 representing survival to hospital discharge.

Admissions were not removed solely because their laboratory data were incomplete. Two admissions had no selected laboratory measurements within the prediction window; both resulted in in-hospital death. These admissions were retained, preserving the dataset's 129 admissions. Missing values were subsequently addressed during model preprocessing rather than excluding these observations during dataset construction.

The final admission-level dataset contained 129 rows and 36 columns. It was used for the subsequent exploratory data analysis, feature engineering and machine learning stages.

2.3 Exploratory Data Analysis

Exploratory data analysis (EDA) was conducted to examine the structure and characteristics of the admission-level dataset before machine learning. The analysis investigated the distribution of the target variable, demographic and admission characteristics, diagnosis categories, laboratory measurements, missing values, potential outliers and correlations between numerical features.

The target variable was examined to assess the balance between surviving and fatal admissions. Of the 129 admissions, 89 (69.0%) resulted in survival to hospital discharge, while 40 (31.0%) resulted in in-hospital death. This imbalance highlighted the importance of evaluating model performance using metrics beyond accuracy alone, particularly recall for the mortality class.

Demographic analysis examined age and gender in relation to the mortality outcome. The median age was higher among admissions resulting in death (79.4 years) than among surviving admissions (72.5 years). However, the distributions overlapped substantially, indicating that age alone did not clearly distinguish between the two groups. The dataset was also concentrated towards older ages, including a concentration at 90 years associated with MIMIC-III's age de-identification procedure. Observed mortality was higher among female than male admissions in this sample, but the small sample size meant that this difference could not be interpreted as evidence of a general difference in underlying mortality risk.

Admission characteristics were examined to understand how patients entered the hospital. **Emergency** admissions accounted for 119 of the 129 observations (92.2%), while **elective** and **urgent** admissions accounted for eight and two observations, respectively. The limited representation of the latter groups restricted meaningful comparisons of mortality across admission types. Admission locations and diagnosis categories were also explored. Mortality rates varied across diagnosis categories, but the small numbers within individual categories made these differences uncertain.

Laboratory measurements were examined through distribution plots and comparisons of median values between surviving and fatal admissions. The mortality group had higher median **glucose, creatinine** and **urea nitrogen** values, alongside slightly lower median **bicarbonate, sodium** and **magnesium** values. Nevertheless, considerable overlap remained between outcome groups. **Creatinine** was right-skewed, with a median of 1.0 and a mean of 1.34. Its maximum value of 10.5 was investigated during data cleaning and retained because it was considered clinically plausible rather than an obvious data-entry error.

Missingness was assessed across the selected laboratory variables. Missing measurements were generally limited, although **calcium** showed the greatest difference between outcome groups, with 11 missing values among surviving admissions and three among fatal admissions. These comparisons were uncertain because of the small sample size. Missing laboratory values were therefore retained and addressed during model preprocessing through imputation.

Finally, a correlation analysis was conducted to investigate relationships between numerical features. The strongest observed correlation was between **haematocrit** and **haemoglobin** ((r=0.89)), indicating that these variables contained substantially overlapping information. **Creatinine** and **urea nitrogen** showed a moderate positive correlation ((r=0.38)), while most other numerical relationships were weak to moderate.

Overall, the EDA identified potentially informative demographic, diagnostic and laboratory features while highlighting limitations that could affect model development and evaluation. These included the small dataset, the imbalance between outcome classes, uneven representation across categorical groups, missing laboratory measurements and correlations between some predictors. The findings informed the subsequent feature engineering and modelling stages, while being treated as exploratory rather than causal or clinically conclusive evidence.

2.4 Feature Engineering

Feature engineering was performed to transform the cleaned admission-level dataset into a format suitable for machine learning. The process involved selecting candidate predictors, preparing categorical variables and defining preprocessing steps for numerical and categorical features.

The candidate predictors included age at admission, gender, diagnosis category and 13 laboratory measurements. The laboratory variables represented a range of clinical measurements, including renal function, electrolyte balance, blood glucose and blood cell characteristics. The target variable, **hospital_expire_flag**, was kept separate from the predictors to define the binary classification task.

Diagnostic information was represented using a grouped primary diagnosis category. Infrequently occurring diagnosis categories were combined into an “**Other**” group when they contained fewer than five observations. This reduced the number of distinct categories and limited the influence of very sparsely represented groups. The number of

recorded diagnoses was explored during feature engineering but was excluded from the final modelling predictors.

Categorical variables, specifically gender and diagnosis category, were converted into numerical features using one-hot encoding. This represents each category as a binary indicator so that the models can process categorical information numerically. The encoding used **drop="first"** to omit one reference category from each categorical variable, while **handle_unknown="ignore"** allowed the preprocessing pipeline to handle categories not encountered during fitting (Pedregosa et al., 2011).

Numerical features were processed through a pipeline that replaced missing values with the median and then applied standardisation. Median imputation allowed admissions with incomplete laboratory measurements to remain available for modelling, while standardisation placed numerical variables on comparable scales. These transformations were incorporated into the model preprocessing workflow so that preprocessing parameters could be learned from the training data rather than the held-out test data.

The resulting features provided a consistent representation of demographic, diagnostic and laboratory information for the classification models. However, the small dataset limited the amount of information available for learning reliable relationships. In addition, because the availability of every diagnosis record within the first 24 hours could not be confirmed, the diagnosis-derived features require caution when interpreting the results as predictions made strictly within that time window.

2.5 Machine Learning Models

Three supervised classification algorithms were investigated to predict in-hospital mortality: **Logistic Regression, Random Forest** and **XGBoost**. These models were selected to compare a relatively simple statistical classifier with two tree-based ensemble approaches. A baseline prediction was also used as a reference point against which model performance could be assessed.

Logistic Regression was included as a comparatively simple classification method that estimates the relationship between predictor variables and the probability of a binary outcome. **Random Forest** was used to investigate whether an ensemble of decision trees could capture more complex relationships between clinical features and mortality (Pedregosa et al., 2011). **XGBoost** was included as a gradient-boosting approach that

builds an ensemble of trees sequentially, with each stage intended to improve predictive performance (Chen and Guestrin, 2016).

The models were developed using the selected demographic, diagnostic and laboratory predictors. A preprocessing pipeline was used to handle missing numerical values through median imputation, standardise numerical features and one-hot encode categorical variables. Combining preprocessing with model fitting helped ensure that transformations were learned within the training data used for each evaluation fold (Pedregosa et al., 2011).

Model performance was assessed using five-fold stratified cross-validation on the training data. Stratification maintained the approximate proportion of surviving and fatal admissions across the folds, while shuffling and a fixed random seed were used to support reproducibility. Accuracy and recall were recorded for each fold. Accuracy measured the overall proportion of correct predictions, while recall measured the proportion of admissions resulting in death that were correctly identified. Recall was particularly important because failing to identify a fatal admission represents a substantial limitation for a mortality prediction model (Pedregosa et al., 2011).

The **Logistic Regression** model was also investigated at different classification thresholds using out-of-fold predicted probabilities. The default threshold of 0.5 was compared with a lower threshold of 0.3 to examine the trade-off between identifying more fatal admissions and generating additional false-positive predictions. This analysis was intended to explore how the choice of threshold affects mortality detection, rather than assuming that the default threshold was necessarily the most appropriate.

Given the small size of the demonstration dataset, the modelling process was treated as an exploratory comparison of classification approaches. The results were interpreted cautiously, with particular attention to variation in performance across cross-validation folds and the ability of each model to identify admissions resulting in death.

2.6 Model Evaluation

Model evaluation was conducted in two stages: cross-validation during model development and a final assessment using a held-out test set. This approach allowed model performance to be compared during development while reserving a separate subset of admissions for final evaluation.

The dataset was divided into training and test sets using a stratified split, with 80% of admissions allocated to training and 20% to testing. A fixed random seed was used to support reproducibility, and stratification preserved the approximate distribution of the mortality outcome across both subsets. The resulting training set contained 103 admissions, while the test set contained 26 admissions.

During model development, five-fold stratified cross-validation was used to compare **Logistic Regression, Random Forest** and **XGBoost**. Accuracy and mortality recall were recorded across the folds to assess both overall classification performance and the ability to identify fatal admissions. The mean and variation in these metrics were considered when comparing the models.

The final evaluation focused on the selected **Logistic Regression** model using the held-out test set. Performance was assessed **using accuracy, precision, recall, F1-score, specificity** and the area under the receiver operating characteristic curve **(ROC-AUC)**. **Accuracy** measured the proportion of correct predictions, while **precision** measured the proportion of predicted deaths that were actual deaths. **Recall** measured the proportion of actual deaths correctly identified, and the **F1-score** summarised the balance between **precision** and **recall**. **Specificity** measured the proportion of surviving admissions correctly identified, while **ROC-AUC** assessed the model's ability to distinguish between the two outcome classes across different classification thresholds (Pedregosa et al., 2011).

A confusion matrix was also examined to distinguish true-positive predictions, true-negative predictions, false positives and false negatives. Particular attention was given to false negatives, representing fatal admissions incorrectly classified as survivors. This was important because overall accuracy could obscure poor detection of the minority mortality class.

The evaluation was interpreted in the context of the dataset's limitations. With only 26 admissions in the held-out test set, including eight deaths, individual predictions could substantially affect the reported metrics. The results were therefore treated as exploratory estimates of performance rather than reliable evidence of clinical effectiveness or generalisability.

3. Results

3.1 Dataset Characteristics

The final dataset contained 129 hospital admissions from 100 unique patients. Each row represented an individual hospital admission, with the binary outcome variable **hospital_expire_flag** indicating whether the admission resulted in in-hospital death. Of the 129 admissions, 89 (69.0%) resulted in survival to hospital discharge, while 40 (31.0%) resulted in death.

The predictor set included **age at admission, gender, primary diagnosis category** and 13 laboratory measurements. These laboratory variables covered several areas of clinical information, including renal function, electrolyte balance, blood glucose and blood cell characteristics. The dataset therefore combined demographic, diagnostic and laboratory information for the subsequent classification experiments.

The dataset was predominantly composed of older patients and emergency admissions. The median age was 72.5 years among surviving admissions and 79.4 years among admissions resulting in death. However, the age distributions overlapped, meaning that age did not clearly separate the two outcome groups. Emergency admissions accounted for 119 observations (92.2%), leaving relatively few elective and urgent admissions for comparison.

Laboratory measurements showed some differences between the outcome groups. Admissions resulting in death had higher median **glucose, creatinine** and **urea nitrogen** values, alongside slightly lower median **bicarbonate, sodium** and **magnesium** values. Nevertheless, the distributions overlapped considerably, and these descriptive differences do not establish that any individual measurement independently predicted mortality.

Missing laboratory values were retained in the dataset and addressed during model preprocessing. Two admissions had no measurements among the selected laboratory variables within the first 24 hours after admission; both were fatal admissions. Retaining these observations preserved the original sample size, but also meant that the modelling pipeline needed to accommodate incomplete predictor information.

Overall, the dataset provided a basis for comparing classification approaches, but its small size and uneven representation of outcome and admission groups limited the strength of conclusions that could be drawn. The observed patterns should therefore be

understood as descriptive findings from a demonstration dataset, rather than generalisable evidence about mortality risk in hospital populations.

3.2 Model Performance

The performance of **Logistic Regression, Random Forest** and **XGBoost** was compared using five-fold stratified cross-validation on the training dataset. Mean accuracy and mortality recall were used to assess overall classification performance and the ability to identify admissions resulting in death.

Logistic Regression achieved a mean cross-validation accuracy of 61.0% and a mean mortality recall of 21.0%. **Random Forest** achieved a slightly higher mean accuracy of 62.1%, but its mean mortality recall was only 3.3%. **XGBoost** performed less effectively, with a mean accuracy of 54.4% and a mean mortality recall of 10.0%. These results indicate that **Random Forest's** marginally higher accuracy did not translate into better identification of fatal admissions. **Logistic Regression** provided the strongest mortality recall of the three models during cross-validation, although its performance remained limited.

Performance also varied between cross-validation folds, particularly for **Logistic Regression**. Its mortality recall ranged from 0% to approximately 42.9%, demonstrating that the model's ability to identify fatal admissions depended substantially on the subset of data used for evaluation. This variability was consistent with the limited number of admissions and mortality events available for model development.

An additional threshold analysis was conducted using out-of-fold predicted probabilities from **Logistic Regression**. At the default classification threshold of 0.50, mortality recall was 22% and precision was 32%. Reducing the threshold to 0.30 increased recall to 44%, while precision remained approximately 32%. At a threshold of 0.20, recall increased further to 56%, while precision decreased to 29%. This demonstrated that lowering the threshold allowed the model to identify a greater proportion of fatal admissions, but also increased the number of admissions predicted to result in death incorrectly.

Overall, **Logistic Regression** was selected for the final held-out test evaluation because it achieved the highest mean mortality recall during cross-validation. However, none of the models demonstrated consistently strong mortality detection. The results therefore suggest that model choice and classification threshold both influenced performance, while the small dataset limited the reliability of comparisons between algorithms.

3.3 Held-Out Test Results

The selected **Logistic Regression** model was evaluated using a held-out test set of 26 hospital admissions, comprising 18 admissions that resulted in survival and eight that resulted in in-hospital death. The model achieved an accuracy of 73.1%, compared with a majority-class baseline accuracy of 69.2%. Although this represented a modest improvement over the baseline, accuracy alone did not adequately reflect the model's ability to identify fatal admissions.

The model achieved a precision of 66.7% and a mortality recall of 25.0%. This meant that, of the eight admissions resulting in death, the model correctly identified two and incorrectly classified the remaining six as survivors. Its **F1-score** was 0.364, reflecting the limited balance between precision and mortality recall. These results indicated that the model missed most fatal admissions despite correctly classifying a majority of the test observations overall.

Specificity was considerably higher at 94.4%, with 17 of the 18 surviving admissions correctly classified. The confusion matrix contained 17 true negatives, one false positive, six false negatives and two true positives. This showed that the model was substantially more effective at identifying surviving admissions than fatal ones. In the context of mortality prediction, the six false-negative predictions were a particularly important limitation.

The model's **ROC-AUC** was 0.507, close to 0.5, which indicates little ability to distinguish between fatal and surviving admissions using its predicted probabilities in this test sample. Taken alongside the low **mortality recall**, this suggests that the model's apparently reasonable accuracy did not correspond to strong predictive discrimination.

These results must be interpreted cautiously because the test set contained only 26 admissions and eight deaths. With so few observations, individual predictions had a substantial effect on the reported metrics. The results therefore provide limited evidence of predictive performance and cannot establish whether the model would generalise to a larger or different patient population. Overall, the held-out evaluation showed that the Logistic Regression model offered only a modest improvement in accuracy over the majority-class baseline and did not reliably identify admissions resulting in death.

4. Discussion

4.1 Interpretation of Findings

The aim of this project was to investigate whether machine learning models could predict in-hospital mortality using demographic, diagnostic and laboratory information associated with the first 24 hours of a hospital admission. The results suggest that this was a challenging prediction task within the available dataset. Although **Logistic Regression** performed better than the other models in terms of mean **mortality recall** during cross-validation, its performance on the held-out test set demonstrated limited ability to identify admissions resulting in death.

The comparison between the three classification models highlighted the importance of selecting evaluation metrics that reflect the intended prediction task. **Random Forest** achieved a slightly higher mean cross-validation accuracy than **Logistic Regression**, at 62.1% compared with 61.0%. However, its mean mortality recall was only 3.3%, compared with 21.0% for **Logistic Regression**. **XGBoost** achieved a mean accuracy of 54.4% and a mean mortality recall of 10.0%. These findings demonstrate that a model can achieve reasonable overall classification accuracy while performing poorly at identifying the outcome of greatest interest. **Logistic Regression** was therefore selected for further evaluation because its comparatively higher mortality recall was more closely aligned with the project's objective.

The held-out test results reinforced this concern. **Logistic Regression** achieved an accuracy of 73.1%, only modestly exceeding the majority-class baseline of 69.2%. However, its mortality recall was 25.0%, meaning that it identified only two of the eight admissions resulting in death. Its specificity of 94.4% showed that it correctly classified most surviving admissions, while its **ROC-AUC** of 0.507 indicated little ability to distinguish between fatal and surviving admissions using predicted probabilities. Taken together, these metrics suggest that the model's accuracy overstated its usefulness for mortality detection when considered in isolation.

The threshold analysis further demonstrated the trade-off between identifying fatal admissions and avoiding false-positive predictions. Reducing the **Logistic Regression** threshold from 0.50 to 0.30 increased out-of-fold **mortality recall** from approximately 22% to 44%, while precision remained around 32%. At a threshold of 0.20, recall increased to 56%, but precision fell to approximately 29%. This illustrates how classification decisions influence the balance between missed deaths and incorrect mortality predictions. However, a lower threshold does not necessarily make a model more useful overall: its suitability depends on the consequences of false-negative and

false-positive predictions, as well as the reliability of the underlying probability estimates.

The exploratory analysis identified differences between surviving and fatal admissions in several demographic and laboratory characteristics. For example, fatal admissions had higher median **age, glucose, creatinine** and **urea nitrogen** values than surviving admissions. These patterns suggest that the selected variables may contain information associated with patient outcomes. Nevertheless, the observed distributions overlapped substantially, and the small sample limited the reliability of comparisons between outcome groups. The exploratory findings therefore cannot establish that any individual variable independently predicts mortality or that the observed associations would remain consistent in a larger population.

Overall, the project demonstrates a complete workflow for preparing clinical data, exploring potential predictors, comparing classification algorithms and evaluating model performance. However, the results do not provide convincing evidence that the models developed in this study could reliably predict in-hospital mortality. The research question was therefore investigated, but the predictive performance achieved was insufficient to support strong conclusions about the practical effectiveness of the approach.

4.2 Limitations

Several limitations affect the interpretation and generalisability of this study. The most significant are the small dataset, the limited number of deaths in the held-out test set, the uncertainty surrounding predictor availability within the intended prediction window, and the restricted scope of the modelling process.

The demonstration dataset was also curated from patients who eventually died, rather than sampled to represent the full hospital population. This selection process further limits the generalisability of the findings to other patient populations and routine clinical settings (Johnson, Pollard and Mark, 2019).

The primary limitation was the size of the MIMIC-III Clinical Database Demo dataset. The analysis included only 129 hospital admissions from 100 unique patients, of which 40 resulted in in-hospital death. Although this dataset was sufficient to demonstrate a machine learning workflow, it provided limited information from which to learn reliable relationships between clinical characteristics and mortality. The small sample also

increased the likelihood that individual observations would influence model performance substantially. Consequently, the results may not reflect the performance that would be obtained using a larger and more representative clinical dataset.

The size of the held-out test set further limited the reliability of the evaluation. The test set contained only 26 admissions, including eight deaths. The model correctly identified two of these deaths and missed the remaining six, resulting in a **mortality recall** of 25.0%. With so few fatal outcomes, correctly or incorrectly classifying even one additional admission would noticeably change the reported metrics. The test results should therefore be interpreted as exploratory estimates rather than precise measures of expected performance in other patient populations.

A further limitation concerns the availability of predictor information at the intended prediction time. The research question specifies prediction using information available during the first 24 hours of admission, and laboratory measurements were selected according to this time window. However, the analysis also included a category derived from primary diagnostic information. The project did not establish that this diagnostic information would always have been available within the first 24 hours. If some diagnostic information was recorded or finalised later, its inclusion could introduce temporal leakage, meaning that the model would use information unavailable at the point when a real prediction was intended to be made. The results therefore cannot conclusively establish the performance of a model operating strictly within the specified 24-hour window.

The dataset also contained missing laboratory measurements. Two admissions had no observations for the selected laboratory variables within the prediction window, and missing values were addressed through median imputation within the preprocessing pipeline. This allowed the admissions to remain in the analysis, but imputation cannot recover the underlying clinical measurements. Missingness may also reflect differences in which tests were ordered or recorded, rather than occurring entirely at random. The study did not establish the reasons for missing measurements, so their potential influence on the models remains uncertain.

The modelling process was exploratory and involved comparing three classification algorithms using a limited set of predictors and largely default model configurations. Although five-fold stratified cross-validation provided a structured basis for comparison, performance varied between folds, particularly for mortality recall. The study did not conduct an extensive hyperparameter search or evaluate the models through repeated validation across multiple random splits. As a result, the analysis cannot establish whether the observed performance represents the best achievable performance from

these algorithms or how stable the model rankings would be under different data partitions.

Finally, the study was limited to a small demonstration dataset and did not include external validation on an independent clinical dataset. The findings cannot therefore be assumed to generalise to other hospitals, patient populations or clinical settings. In addition, the analysis investigated predictive associations rather than causal relationships. Differences observed between surviving and fatal admissions do not demonstrate that individual features caused the outcomes.

Taken together, these limitations mean that the models should be regarded as proof-of-concept implementations rather than clinically reliable mortality prediction tools. The findings are useful for demonstrating the healthcare data science workflow and identifying areas for improvement, but stronger conclusions would require a larger dataset, careful verification of predictor timing, more extensive model development and independent validation.

4.3 Future Improvements

Several improvements could strengthen the reliability of the analysis and provide a more robust assessment of machine learning for in-hospital mortality prediction.

The most important improvement would be to increase the size of the dataset. The MIMIC-III Clinical Database Demo was sufficient to demonstrate the modelling workflow, but its small number of admissions limited the amount of information available for training and evaluation. Future work could use a larger clinical dataset, subject to the relevant access requirements, to provide more examples of both surviving and fatal admissions. A larger dataset would allow the models to learn from a wider range of clinical presentations and support more reliable estimates of predictive performance.

A second improvement would be to enforce a stricter definition of the prediction window. All predictors should be verified as available at the intended prediction time, rather than simply being associated with the same hospital admission. In particular, the timing of the primary diagnosis information should be investigated. If its availability within the first 24 hours cannot be established, the diagnosis-derived feature should be excluded from a revised model. This would reduce the risk of temporal data leakage and ensure that the evaluation more closely reflects the intended prediction task.

The data-splitting strategy could also be improved by accounting for repeated admissions from the same patient. The current analysis divided the data at admission level, even though the dataset contained 129 admissions from 100 unique patients. Consequently, admissions belonging to the same individual could potentially appear in both the training and test sets. Future work should use a patient-level split, ensuring that all admissions belonging to a given patient are assigned to the same subset. This would provide a stronger assessment of how well the model generalises to patients whose admissions were not represented during training.

Model development could be strengthened through more extensive validation and systematic hyperparameter tuning. Repeated stratified cross-validation could help assess the stability of model performance across different data partitions, while a structured search of model parameters could investigate whether alternative configurations improve predictive performance. These steps would be more informative with a larger dataset, as the current demonstration sample provides limited scope for reliable model optimisation. Performance estimates should also be accompanied by uncertainty measures, such as confidence intervals where appropriate, to make the limitations of the available evidence clearer.

Future evaluation should continue to prioritise **mortality recall** alongside **precision**, **specificity**, **F1-score** and **ROC-AUC**. The threshold analysis demonstrated that lowering the classification threshold increased mortality recall but also affected precision. Further work could examine this trade-off more systematically and assess the calibration of predicted probabilities. In a clinical setting, the choice of threshold would need to reflect the intended use of the model and the relative consequences of missed deaths and false-positive predictions. A threshold should therefore not be selected solely because it produces the highest recall.

Finally, any model developed using a larger dataset would require independent validation before its results could be considered generalisable to other clinical settings. External validation on data from a separate patient population or hospital would help establish whether performance was maintained beyond the development dataset. Further evaluation would also be needed to investigate potential differences in performance across patient groups and to consider the clinical, ethical and practical implications of using mortality predictions to support decision-making.

These improvements would build on the workflow established in this project while addressing its main methodological limitations. The immediate priority would be to strengthen the dataset and prediction-window definition, followed by more robust validation and model development. Clinical deployment would remain outside the scope

of the current project and would require substantially stronger evidence of reliability, generalisability and clinical usefulness.

5. Conclusion

This project investigated whether machine learning could predict in-hospital mortality using demographic, diagnostic and laboratory information associated with the first 24 hours of a hospital admission. Using the MIMIC-III Clinical Database Demo, an admission-level dataset of 129 hospital admissions was constructed and used to compare **Logistic Regression**, Random Forest and XGBoost classification models. The project followed a structured workflow encompassing data cleaning, exploratory data analysis, feature engineering, model development and evaluation.

Logistic Regression achieved the highest mean mortality recall during five-fold stratified cross-validation, although its performance varied considerably between folds. In the held-out test evaluation, the model achieved 73.1% **accuracy** but identified only two of the eight admissions resulting in death, corresponding to a **mortality recall** of 25.0%. Its **ROC-AUC** of 0.507 also indicated little ability to distinguish between fatal and surviving admissions in the test sample. These findings demonstrate that **accuracy** alone can provide a misleading impression of model performance when the objective is to identify a specific clinical outcome.

The results do not provide convincing evidence that the models developed in this project could reliably predict in-hospital mortality. The small dataset, limited test sample and uncertainty surrounding the availability of diagnostic information within the intended prediction window restrict the conclusions that can be drawn. The models should therefore be regarded as proof-of-concept implementations rather than clinically reliable prediction tools.

Despite these limitations, the project demonstrates the application of a complete healthcare data science workflow to a clinical prediction problem. It also highlights the importance of appropriate evaluation metrics, careful feature selection, validation strategies and critical interpretation of model performance. Future work using a larger dataset, stricter control of predictor timing, patient-level data splitting and more robust validation could provide a stronger assessment of the feasibility of this approach.

In conclusion, the project established a foundation for investigating machine learning in healthcare analytics, but the predictive performance achieved was insufficient to support reliable mortality prediction. Its principal contribution was demonstrating the analytical process and identifying the methodological improvements required for a more robust investigation.

6. References

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